SMART FERTILITY TOOLS

A 2025 *Nature Scientific Reports* study found the MAIA (Morphological Artificial Intelligence Assistance) platform—an AI model designed to assist embryologists in assessing and selecting an embryo from several for plantation during IVF—achieved successful implantation prediction rates of 70.1 per cent, outperforming traditional assessments by around five percentage points. Such grading of embryos by MAIA and ERICA (Embryo Ranking Intelligent Classification Algorithm), deep-learning systems that provide real-time embryo rankings based on predicted implantation potential, allows clinicians to transfer embryos with greater confidence.

"AI is being used to support different stages of IVF. It brings in precision, consistency and additional insights that help doctors and embryologists make better decisions," says Dr Prachi Benara, fertility specialist at Birla Fertility & IVF, Gurugram. AI-assisted tools, she explains, can examine embryo images and growth patterns, identifying the ones that hold the best chance of successful implants. "It adds an extra layer of accuracy to the traditional selection process, which is usually based on visual observation."

At Nova IVF fertility clinics across India, the introduction of Vita Embryo—Kai Health's AI-powered embryo assessment tool—has added precision. "It identifies subtle details in embryo health that the human eye may miss. Clinical data shows AI-powered embryo selection can increase precision by 12 per cent compared to human evaluation," says Dr Sujatha Ramakrishnan, head of embryology at Nova. Vita Embryo tools are being deployed at many clinics, including DSS Imagetech's IVF vertical Hanahealth.

Indeed, AI is making a difference right at the beginning—in the selection of oocytes, or egg cells, the quality of which strongly influences fertilisation success. Embryologists usually rely on visual assessment of an egg's size, symmetry and surface structure. But these judgements are subjective. Computer vision systems using AI can detect subtle features that escape the eye, improving outcomes. A 2024 study in *Human Reproduction* showed fertilisation rates rising from 83.2 per cent under manual evaluation to 92.6 per cent with AI-assisted scoring.

"We need more published data comparing AI-based outcomes with those from embryologists. Right now, there are discrepancies. High costs, patchy regulation and limited evidence are holding AI back"

A.R. GHATAK
Vice president, IVF division, Hanahealth

Sperm screening is the other introductory process. "AI tools can evaluate sperm motility, shape and overall health more precisely than the human eye," says Benara. "By identifying the healthiest sperm for fertilisation, it improves the chances of creating strong embryos." Dr Gauri Agarwal, founder of Seeds of Innocence, a chain of IVF centres, adds that AI-powered microscopes can automatically measure sperm count and morphology.

Researchers are also developing non-invasive AI tools that detect chromosomal abnormalities such as aneuploidy—a genetic disorder resulting in extra or missing chromosomes, a significant cause of miscarriages—by analysing embryo images. "Traditionally, pre-implantation genetic testing for aneuploidy has been invasive. With AI, embryologists can now assess embryos non-invasively and often with greater precision," says Manjunath.

Even the timing of oocyte retrieval—the optimal period of retrieving eggs after ovulation for fertilisation—is being refined. ILETIA, an AI model, predicts the optimal interval between the ovulation trigger injection (a hormone shot that triggers the release of eggs in IVF) and egg collection.

The impact of these advances is best understood through lived experience. Navjot Kaur, a 37-year-old homemaker from Mumbai, was long sceptical about IVF. When she chose a clinic using AI for oocyte selection, embryo grading, sperm analysis and optimised retrieval timing, her outlook shifted. Each stage was explained with data. AI amplified—not replaced—the human care she received, and she was recently blessed with a daughter. Indeed, doctors stress that AI is not a replacement for human experience. Embryologists' contextual judgement remains crucial, particularly in complex or borderline cases.

PROMISE AND REALITY

For Indian couples, IVF is an expensive path. A standard cycle (the entire treatment lasts for 3-6 cycles) costs between Rs 1 lakh and Rs 4 lakh . With AI tools—specialised imaging systems, data platforms, licensing and training—the bill rises by another 20–50 per cent. That can push treatment to nearly Rs 6 lakh per cycle.

With her modest savings, Aisha Singh, a 33-year-old schoolteacher from Mumbai, could barely afford one IVF attempt. Paying extra for AI-aided treatment seemed impossible. "Our polluted environment makes infertility a public health issue," she says. Her frustration reflects a growing demand for policies that see IVF as essential care, not an elective choice.